Sam

For my Heart Disease Prediction & Signal Analysis midterm project, I designed and built the entire 16-section experimental workflow from raw data preprocessing all the way through model explainability, feature engineering, and controlled signal verification. I started by cleaning the 10,000-patient dataset—imputing missing numerical values with the median and categorical values with the mode—and then implemented a leak-free preprocessing pipeline where I encoded categorical features and split the raw data before fitting StandardScaler strictly on the training set. When my baseline Random Forest model failed to predict the minority class (0.00 recall on positive cases), I personally drove the investigation across multiple imbalance and optimization techniques: I computed balanced class weights, wrote custom Precision-Recall curve logic to test optimal (0.0600) and balanced (0.5000) decision thresholds, applied SMOTE oversampling, and trained an XGBoost classifier with scale_pos_weight alongside 5-fold cross-validation and GridSearchCV hyperparameter tuning. Rather than guessing why the models stayed around a ~0.50 ROC-AUC, I contributed a full diagnostic and explainability section using Pearson correlations, permutation importance, and SHAP TreeExplainer (bar, beeswarm, and dependence plots) to mathematically prove that the raw synthetic target was generated independently of the 20 patient features (all linear correlations within ±0.02, with Stress Level at just 0.016). To push the feature space further, I engineered degree-2 polynomial interactions across key biomarkers, applied SelectKBest and RFECV feature selection, consolidated redundant lipid and blood pressure columns, and created three domain-specific clinical indices from scratch: a Metabolic Risk Score (BMI * Triglyceride Level * Fasting Blood Sugar), a Cardiovascular Strain Index (Blood Pressure * Age), and an Inflammatory-Vascular Score (CRP Level * Homocysteine Level). Finally, to prove that my machine learning architecture was sound, I designed a custom clinical signal verification experiment weighted by real cardiovascular risk factors (Age, BMI, Blood Pressure, Cholesterol Level, Homocysteine Level, Smoking, and Diabetes), added stratified shuffling, cleaned feature names for LightGBM, and built both a weighted Soft-Voting Ensemble (XGBoost + LightGBM + Random Forest) that achieved a 0.9948 ROC-AUC and a final high-capacity Logistic Regression model that reached a 1.0000 ROC-AUC with 100% accuracy, precision, and recall, wrapping the notebook up with a complete comparative summary table in Section 16.

Ebuka

For my contribution to this project, I worked on preparing the Heart Attack Prediction dataset for machine learning. I started with the dataset from Kaggle and reviewed its columns to understand what information was available and what we could try to predict. I helped organize the project around predicting heart attack risk using factors such as age, cholesterol, blood pressure, and lifestyle information. I worked with the Jupyter Notebook and Python code to split the dataset into training and testing sets, check for missing values, process categorical and numerical features, and balance the training data. I also included Matplotlib graphs to compare the data before and after preprocessing and made sure the processed data could be exported as a separate CSV file. Another part of my contribution was putting together the project proposal and organizing the files needed for submission, including the dataset source, notebook, and cleaned

dataset. Working on this helped me get more practice with the data preparation process and understand why each step matters. I also learned the importance of checking the results instead of assuming the code is doing everything correctly. Overall, I contributed to preparing the dataset and organizing the project so it can be used for building and evaluating machine learning models.

Cersidy

My contribution focused on building a model from training the heart-diseases.csv, I applied an ensemble algorithm of classification type to predict the probability of a patient having heart-disease. The classification task by predicting categorical outcomes. Through this process, I gained hands-on experience in selecting appropriate algorithms, training models, and evaluating their performance. This module strengthened my understanding of supervised machine learning and helped me appreciate how different algorithms are applied to solve real-world prediction problems.

This is another student's contribution journal, focused on training a classification model using an ensemble algorithm, whereas your contribution journal focuses on dataset preparation, preprocessing, and organizing the project deliverables.

Sadia Ravate

## 1. Individual Contributions

I contributed to the midterm project by working with the heart disease dataset and setting up the analysis in Google Colab. I worked with the `heart_disease.csv` file and confirmed that it loaded successfully into the notebook.

I examined the dataset using Python and pandas to understand its structure, including its rows, columns, and missing values. This helped me identify the data preparation steps needed before the dataset could be used for machine learning.

I also reviewed the notebook's preprocessing workflow, including missing-value handling, categorical encoding, and numerical scaling. I worked on understanding how these techniques prepare the dataset for analysis and how they relate to the overall goal of predicting heart disease status.

## 2. Tools and Technologies Used

The tools and technologies involved in my work included:

- **Google Colab:** To open and run the Jupyter notebook in a cloud environment.
- **Python:** To perform data analysis and preprocessing.

- **Pandas:** To load, inspect, and manipulate the dataset.
- **Scikit-learn:** For preprocessing techniques and machine learning workflows included in the notebook.
- **Matplotlib:** For creating visualizations to help interpret the data.

## 3. Skills and Knowledge Gained

Through my contributions, I strengthened my understanding of loading datasets, inspecting data quality, identifying missing values, and preparing data for machine learning. I also learned why training and testing data must be separated correctly and why preprocessing should be fitted using training data to prevent data leakage.

This work helped me better understand the relationship between data quality and model performance. It also reinforced the importance of documenting the steps taken so that the project can be reviewed and reproduced.

## 4. Remaining Work and Next Steps

The remaining work includes verifying that the notebook follows the required 70% training and 30% testing split, ensuring that exploratory data analysis is performed using the training data only, reviewing the before-and-after preprocessing visualizations, and exporting the final cleaned dataset.

The final notebook and supporting documents must also be organized in the project's GitHub repository before submission.

## 5. Summary

My individual contribution focused on setting up the dataset in Google Colab, examining its structure and data quality, and reviewing the notebook's preprocessing workflow. This experience helped me apply concepts from class to a practical machine learning project and identify the steps needed to prepare a reliable final submission.